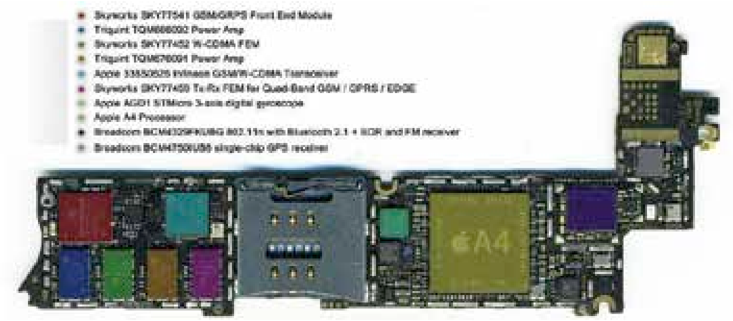

Sensors are an important part of an embedded system - shown here is the iPhone4 PCB

is an important consideration for systems which need to support a change in any of those sub-components. Consider the example of a microwave oven – if the temperature sensor starts working erratically, then one might need to change just the temperature sensor instead of changing the entire control panel. In such cases, the number of components and the option of upgradability must be thought out well ahead of production.

Components in any system (even non-embedded systems) serve as a window of interaction between the system and the outer environment. Taking the microwave oven example again, the controller wouldn't be able to know the temperature inside the heating cabinet if it were not for the temperature sensors. Components are connected to the main system via interfaces. Depending on the need, the interface can be either technology dependent or technology independent. A technology dependent interface connects the component to the board; this interface is not widely used and serves the purpose of controlling, programming and debugging the component alone. A technology independent interface can control the overall functionality of the component, and can send and receive data.

Networking and communication

While embedded systems which are used as networking devices must ensure networking between interfaces, other programmable electronic systems which use multiple components often have to do that as well. An embedded system must ensure that different components which are connected to the system are able to communicate with each other. One of the best examples in our day-to-day lives of this feature is Wi-Fi hotspotting available in smartphones: the phones can receive and send data to the network service